

root@Sriram: ~/xv6-public

root@Sriram:~/xv6-public# make clean

rm -f *.tex *.dvi *.idx *.aux *.log *.ind *.ilg \

*.o *.d *.asm *.sym vectors.S bootblock entryother \

initcode initcode.out kernel xv6.img fs.img kernelmemfs \

xv6memfs.img mkfs .gdbinit \

_cat _echo _forktest _grep _init _kill _ln _ls _mkdir _rm _sh _stressfs _usertests _wc _zombie _hello _sleep

root@Sriram:~/xv6-public#

```

root@Sriram: ~/xv6-public
root@Sriram:~/xv6-public# cd xv6-public/
root@Sriram:~/xv6-public# make clean
rm -f *.tex *.dvi *.idx *.aux *.log *.ind *.ilg \
*.o *.d *.asm *.sym vectors.S bootblock entryother \
initcode initcode.out kernel xv6.img fs.img kernelmemfs \
xv6memfs.img mkfs .gdbinit \
_cat _echo _forktest _grep _init _kill _ln _ls _mkdir _rm _sh _stressfs _usertests _wc _zombie _hello _sleep
root@Sriram:~/xv6-public# make
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -fno-pic -O -nostdinc -I. -c bootmain.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -fno-pic -nostdinc -I. -c bootasm.S
ld -m elf_i386 -N -e start -Ttext 0x7C00 -o bootblock.o bootasm.o bootmain.o
objdump -S bootblock.o > bootblock.asm
objcopy -S -O binary -j .text bootblock.o bootblock
./sign.pl bootblock
boot block is 451 bytes (max 510)
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o bio.o bio.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o console.o console.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o exec.o exec.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o file.o file.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o fs.o fs.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o ide.o ide.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o ioapic.o ioapic.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o kalloc.o kalloc.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o kbd.o kbd.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o lapic.o lapic.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o log.o log.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o main.o main.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o mp.o mp.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o picirq.o picirq.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o pipe.o pipe.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o proc.o proc.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o sleeplock.o sleeplock.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o spinlock.o spinlock.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o string.o string.c

```



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root@Sriram: ~/xv6-public
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o syscall.o syscall.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o sysfile.o sysfile.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o sysproc.o sysproc.c
gcc -m32 -gdwarf-2 -Wa,-divide -c -o trapasm.o trapasm.S
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o trap.o trap.c
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o uart.o uart.c
./vectors.pl > vectors.S
gcc -m32 -gdwarf-2 -Wa,-divide -c -o vectors.o vectors.S
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o vm.o vm.c
gcc -m32 -gdwarf-2 -Wa,-divide -c -o entry.o entry.S
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -fno-pic -nostdinc -I. -c entryother.S
ld -m elf_i386 -N -e start -Ttext 0x7000 -o bootblockother.o entryother.o
objcopy -S -O binary -j .text bootblockother.o entryother
objdump -S bootblockother.o > entryother.asm
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -nostdinc -I. -c initcode.S
ld -m elf_i386 -N -e start -Ttext 0 -o initcode.out initcode.o
objcopy -S -O binary initcode.out initcode
objdump -S initcode.o > initcode.asm
ld -m elf_i386 -T kernel.ld -o kernel entry.o bio.o console.o exec.o file.o fs.o ide.o ioapic.o kalloc.o kbd.o lapic.o log.o main.o mp.o picirq.o pipe.o proc.o sleeplock.o spinlock.o string.o swtch.o syscall.o sysfile.o sysproc.o trapasm.o trap.o uart.o vectors.o vm.o -b binary initcode entryother
objdump -S kernel > kernel.asm
objdump -t kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel.sym
dd if=/dev/zero of=xv6.img count=10000
10000+0 records in
10000+0 records out
5120000 bytes (5.1 MB, 4.9 MiB) copied, 0.0496431 s, 103 MB/s
dd if=bootblock of=xv6.img conv=notrunc
1+0 records in
1+0 records out
512 bytes copied, 0.0001334 s, 3.8 MB/s
dd if=kernel of=xv6.img seek=1 conv=notrunc
394+1 records in
394+1 records out
201784 bytes (202 kB, 197 KiB) copied, 0.002045 s, 98.7 MB/s
root@Sriram:~/xv6-public#

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objdump -S _usertests > usertests.asm
objdump -t _usertests | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > usertests.sym
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -c usertests.c -o usertests.o
ld -m elf_i386 -N -e main -Ttext 0 -o _wc wc.o ulib.o usys.o printf.o umalloc.o
objdump -S _wc > wc.asm
objdump -t _wc | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > wc.sym
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -c wc.c -o wc.o
ld -m elf_i386 -N -e main -Ttext 0 -o _zombie zombie.o ulib.o usys.o printf.o umalloc.o
objdump -S _zombie > zombie.asm
objdump -t _zombie | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > zombie.sym
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -c zombie.c -o zombie.o
ld -m elf_i386 -N -e main -Ttext 0 -o _hello hello.o ulib.o usys.o printf.o umalloc.o
objdump -S _hello > hello.asm
objdump -t _hello | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > hello.sym
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -c hello.c -o hello.o
ld -m elf_i386 -N -e main -Ttext 0 -o _sleep sleep.o ulib.o usys.o printf.o umalloc.o
objdump -S _sleep > sleep.asm
objdump -t _sleep | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > sleep.sym
./mkfs fs.img README _cat _echo _forktest _grep _init _kill _ln _ls _mkdir _rm _sh _stressfs _usertests _wc _zombie _hello _sleep
nmeta 59 (boot, super, log blocks 30 inode blocks 26, bitmap blocks 1) blocks 941 total 1000
ballocc: first 702 blocks have been allocated
ballocc: write bitmap block at sector 58
qemu-system-i386 -serial mon:stdio -drive file=fs.img,index=1,media=disk,format=raw -drive file=xv6.img,index=0,media=disk,format=raw -smp 2 -m 512
xv6...
cpu0: starting 0
sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart 2 inodestart 32 bmap start 58
init: starting sh
$ hello
Hello Xv6!
Hello from the Kernel!
$ _

```

QEMU - Press Ctrl+Alt+G to release grab

Machine View

SeaBIOS (version 1.15.0-1)

iPXE (https://ipxe.org) 00:03.0 CA00 PCI2.10 PnP PMM+1FF8B590+1FECB590 CA00

Booting from Hard Disk...

cpu0: starting 0

sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart 2 inodestart 32 bmap start 58

init: starting sh

\$ hello

Hello Xv6!

Hello from the Kernel!

\$

```
$ la
exec: fail
exec la failed
$ ls
README      2  2  2286
cat          2  3 15464
echo         2  4 14340
forktest    2  5  8792
grep         2  6 18308
init         2  7 14964
kill         2  8 14428
ln           2  9 14328
ls           2 10 17468
mkdir        2 11 14448
rm           2 12 14432
sh           2 13 28488
stressfs     2 14 15360
usertests    2 15 62860
wc           2 16 15884
zombie       2 17 14012
hello        2 18 14180
sleep        2 19 14468
console      3 20  0
```

```
$ _
```

QEMU - Press Ctrl+Alt+G to release grab

Machine View

ls	2 10 17468
mkdir	2 11 14448
rm	2 12 14432
sh	2 13 28488
stressfs	2 14 15360
usertests	2 15 62860
wc	2 16 15884
zombie	2 17 14012
hello	2 18 14180
sleep	2 19 14468
console	3 20 0
\$	

```
. 1 1 512
/
.. 1 1 512
/
README 2 2 2286
cat 2 3 15464
echo 2 4 14340
forktest 2 5 8792
grep 2 6 18308
init 2 7 14964
kill 2 8 14428
ln 2 9 14328
ls 2 10 17468
mkdir 2 11 14448
rm 2 12 14432
sh 2 13 28488
stressfs 2 14 15360
usertests 2 15 62860
wc 2 16 15884
zombie 2 17 14012
hello 2 18 14180
sleep 2 19 14468
console 3 20 0
```

\$ _

QEMU - Press Ctrl+Alt+G to release grab

Machine View

ls	2 10 17468
mkdir	2 11 14448
rm	2 12 14432
sh	2 13 28488
stressfs	2 14 15360
usertests	2 15 62860
wc	2 16 15884
zombie	2 17 14012
hello	2 18 14180
sleep	2 19 14468
console	3 20 0
\$	

```
/
README      2 2 2286

cat         2 3 15464

echo       2 4 14340

forktest   2 5 8792

grep       2 6 18308

init       2 7 14964

kill       2 8 14428

ln         2 9 14328

ls         2 10 17468

mkdir      2 11 14448

rm         2 12 14432

sh         2 13 28488

stressfs   2 14 15360

usertests  2 15 62860

wc         2 16 15884

zombie     2 17 14012

hello     2 18 14180

sleep     2 19 14468

console    3 20 0
```

```
$ sleep
Usage : Sleep <ticks>
$ sleep 300
```

QEMU - Press Ctrl+Alt+G to release grab

Machine View

mkdir	2	11	14448
rm	2	12	14432
sh	2	13	28488
stressfs	2	14	15360
usertests	2	15	62860
wc	2	16	15884
zombie	2	17	14012
hello	2	18	14180
sleep	2	19	14468
console	3	20	0

```
$ sleep
Usage : Sleep <ticks>
$ sleep 300
$ _
```